

Project Design Phase Solution Architecture

Date	26 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Solution Architecture:

The Human Development Index (HDI) Predictor follows a multi-layer architecture consisting of the User Layer, Frontend Layer, Flask Backend Layer, Data Processing Layer, Machine Learning Prediction Engine, Model Storage Layer, and Deployment Layer.

Users enter country development indicators such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI Per Capita through a web interface. The Flask backend validates the input, preprocesses the data, loads the trained Linear Regression model, predicts the HDI score, classifies it into Very High, High, Medium, or Low Human Development, and displays the result instantly through the web application.

Example - Solution Architecture Diagram:

